Modern Physics

Lecture 5: Quantum Mechanics and Electron Spin

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Books: *Ch 7 of* Arthur Beiser,
Concepts of Modern Physics
Ch 41.5-41.7 of Hugh D. Young and Roger A. Freedman, University Physics with Modern Physics

Recap: The first lecture

How does your phone receive a signal even though nothing is connecting it to the tower?





How does solar calculator work?

Recap: The first lecture

How does your phone receive a signal even though nothing is connecting it to the tower?

The oscillating electric field in radio waves drives the collective motion of charges in the antenna.

Electromagnetic Wave

How does solar calculator work?

Photon incident on semiconductor, electron promoted across band gap.

ELECTRON-PHOTON interaction

How do I know which description to use

Did the underlying physics change?
Or did the relevant physical scale change?

Before describing or designing a system, ask:

What are the characteristic length and energy scales of the system?

Compare the scale of the experiment/device with the characteristic scales of the particles and interactions involved.

The comparison of scales tells us which physical effects can become important.

What are the relevant scales?

The comparison of scales tells us which physical effects can become important.

$$L \sim \lambda_{DB}$$

Quantum effects can become important.

Back to Diffraction

One photon → one localized hit.

Many photons, one at a time → diffraction pattern

Q: Can we predict where the next particle will hit?

Q: Can we predict how many particles will hit different regions after many repetitions?

Q: What exactly does the pattern represent in the particle picture?

Back to Diffraction

One photon → one localized hit.

Many photons, one at a time → diffraction pattern

Q: Can we predict where the next particle will hit?

NO

Q: Can we predict how many particles will hit different regions after many repetitions?

Yes

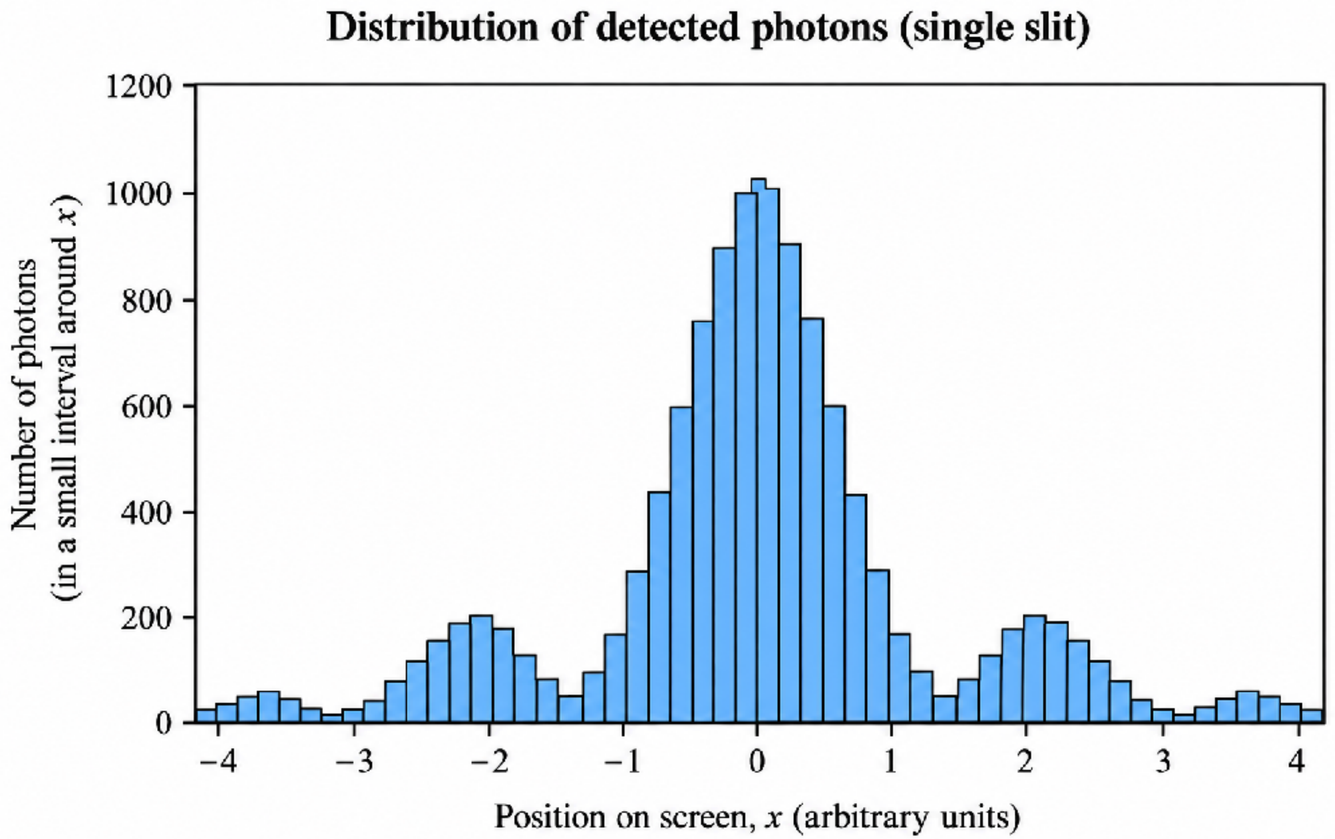
Q: What exactly does the pattern represent in the particle picture?

Probability of detecting the particle at different positions

Back to Diffraction

What kind of prediction is nature allowing us to make?

A probability distribution



How to describe quantum object?

Classical mechanics

If I know

$$\mathbf{r}(0), \qquad \mathbf{p}(0),$$

Newton's equation gives

$$\mathbf{r}(t)$$

—the trajectory.

Our diffraction experiment

Same preparation.
Different individual detection positions.

Can a definite trajectory still be the fundamental description?

How to describe quantum object?

And uncertainty principle told us the same thing

$$\Delta y \, \Delta p_y \geq \frac{\hbar}{2} .$$

Then,

Can we specify simultaneously the exact position and exact momentum needed to construct a classical trajectory?

No.

Our experiment is telling us that a definite Newtonian trajectory is no longer the useful object for describing what we observe.

So now you have **two independent clues**:

Diffraction	Uncertainty
⇓	⇓
Probabilistic outcomes	No exact (y, p_y)

The Wavefunction

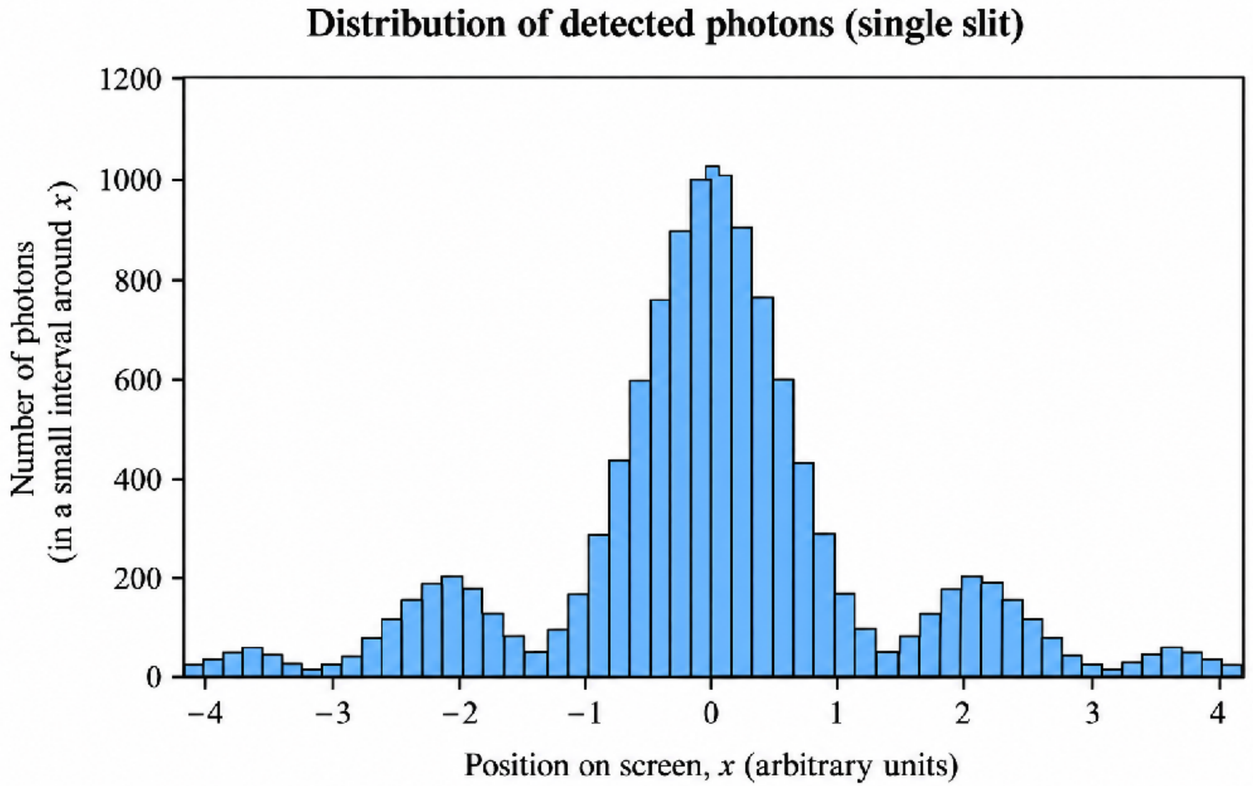
What should replace a trajectory?

We need something that contains enough information to predict the probabilities of different measurement outcomes.

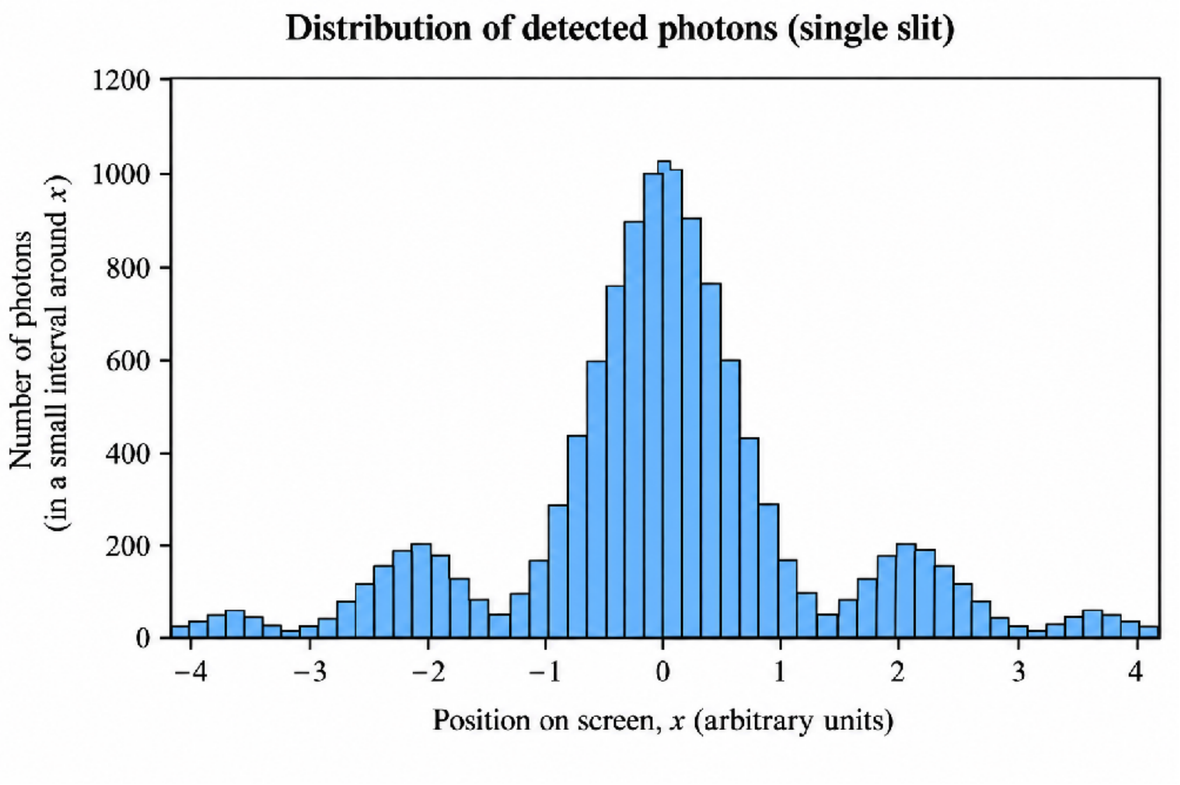
The wave function

$$\psi(\mathbf{r}, t)$$

The **wave function** describes the quantum state.



Wavefunction and Diffraction



The theory does not assign one definite Newtonian trajectory and then hide it from us. It evolves a quantum state, from which probabilities of measurement outcomes are obtained.

Such that

$|\psi(\mathbf{r}, t)|^2 =$ probability density for finding the particle at $\mathbf{r}$

Or

$|\psi(y)|^2 dy = P(\text{particle detected between } y \text{ and } y + dy)$

Evolution of Wavefucntion with time

Newtonian mechanics has an equation of motion.

What is the corresponding governing equation for ψ?

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$$

Schrödinger equation

Here, $\hat{H}$ is Hamiltonian (an operator).

QM puts measurable quantities at the centre

Atomic spectroscopy showed that atoms make transitions between **discrete energy states**.

Heisenberg developed a description based directly on these observable transitions, rather than imagining a definite electron trajectory.

The mathematical quantities that emerged did **not behave like ordinary numbers** - the order of multiplication could matter.

The simplest familiar objects with this property are **matrices**:

$$AB \neq BA\,.$$

Quantum mechanics generalizes this idea:

Measurable physical quantities $\longrightarrow$ **Operators**

$$x \to \hat{x}, \qquad p \to \hat{p}, \qquad E \to \hat{H}\,.$$

An operator needs something to act on.

$$\hat{H}\,\psi$$

Just as a matrix acts on a vector, an operator acts on a **quantum state**. **Hamiltonian operator is associated with the total energy of the system.**

Classical Analogy

Suppose you specify the force law

$$V(r) = -\frac{GMm}{r},$$

↓

and solve Newton's equation

$$\mathbf{F} = -\nabla V = m\mathbf{a}$$

↓

The equations tell you what kinds of trajectories are possible:

circle, ellipse etc.

↓

Then the initial conditions

$\mathbf{r}(0)$, $\mathbf{p}(0)$

determine the particular trajectory the object follows.

Recap: Quantum Mechanics

Specify the physical system

$V(r) = 0$ **or** $V(r) = -\dfrac{e^2}{4\pi\varepsilon_0 r}$

↓

Solve the time-independent Schrodinger equation $\hat{H}\psi = E\psi$

↓

Find the allowed states and physical quantities like energy

$\psi_n, \quad E_n, \quad L^2, \quad L_z, \ldots$

↓

Specifying the initial quantum state $\psi(\mathbf{r},0)$

↓

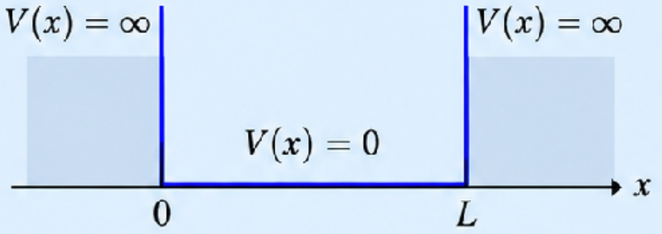
determine its subsequent evolution and measurement probabilities

$\psi(\mathbf{r}, t) \quad |\psi(\mathbf{r}, t)|^2$

Recap from Prof. Vandana's lecture

1. Specify the physical system

$V(r) =$ particle in $1d$ box



2. Solve the time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

For the particle in a 1D box ($0 < x < L$):

$$V(x) = 0 \;\Rightarrow\; -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0, \quad k^2 = \frac{2mE}{\hbar^2}$$

3. Find the allowed states and physical quantities like energy

Apply boundary conditions for an infinite well:

$$\psi(0) = 0, \quad \psi(L) = 0$$

$$\Rightarrow \; k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \ldots$$

Allowed states and energies:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \ldots$$

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots$$

Hydrogen Atom

For the hydrogen atom, the electron experiences the Coulomb potential

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$$

$\downarrow$

We solve the time-independent Schrodinger equation

$$\hat{H}\psi = E\psi$$

$\downarrow$

We obtain the allowed wave functions for the electron

$$\psi_{nlm_l}(r, \theta, \phi)$$

$\downarrow$

and the corresponding allowed energy levels

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

$$n = 1,2,3,\ldots$$

Angular Momentum

The solution gives two additional quantized quantities

$$L^2 = l(l+1)\hbar^2, \qquad l = 0,1,2,\ldots,n-1$$

$$L_z = m_l\hbar, \qquad m_l = -l,-l+1,\ldots,0,\ldots,l-1,l$$

Why did additional quantum numbers appear?

$$V(r) = V(|\mathbf{r}|)$$

The hydrogen potential is spherically symmetric

and remains unchanged under rotations.

$n \longrightarrow$ **Energy**

$l \longrightarrow$ **Magnitude of orbital angular momentum**

$m_l \longrightarrow$ **Component of orbital angular momentum along a chosen axis**

What information appeared naturally from the allowed wave functions?

$n, \qquad l, \qquad m_l$

Fine Structure

Unperturbed Hamiltonian (hydrogen atom)

$$H = \frac{p^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r}$$

Solve the Schrödinger equation

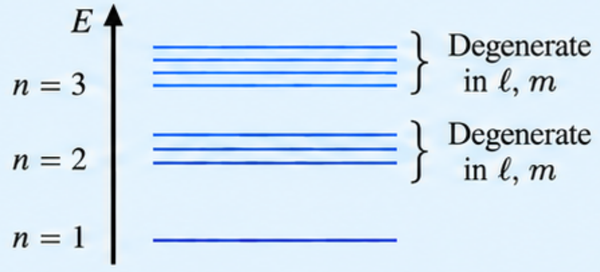
$$H\psi_{n\ell m} = E_n \psi_{n\ell m}$$

Eigenstates and energies

$$\psi_{n\ell m}(r, \theta, \phi)$$

$$E_n = -\frac{me^4}{2(4\pi\epsilon_0)^2\hbar^2}\frac{1}{n^2}$$

Energy depends only on n (all ℓ, m states with the same n are degenerate).



Modified Hamiltonian

$$H' = H + H_{\text{extra}}$$

We add small relativistic and spin-dependent corrections to the Hamiltonian:

$$H_{\text{extra}} = H_{\text{rel}} + H_{\text{SO}} + H_{\text{Darwin}}$$

- relativistic kinetic energy correction
- spin–orbit coupling
- Darwin term

$$H_{\text{extra}} \ll H$$

(treated as a small correction)

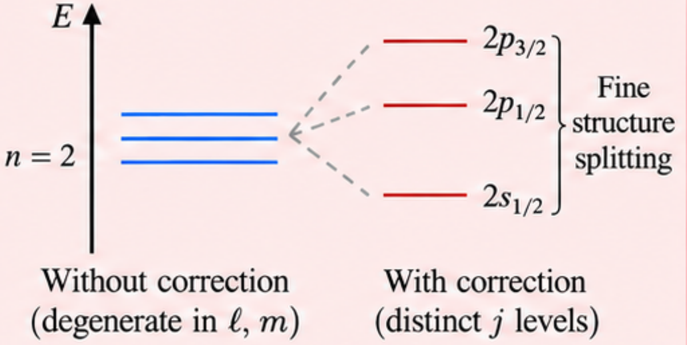
New eigenstates and energies

$$H'\psi'_{n\ell jm} = E'_{nj}\psi'_{n\ell jm}$$

$$\psi'_{n\ell jm} = \psi_{n\ell m} + \text{small corrections}$$

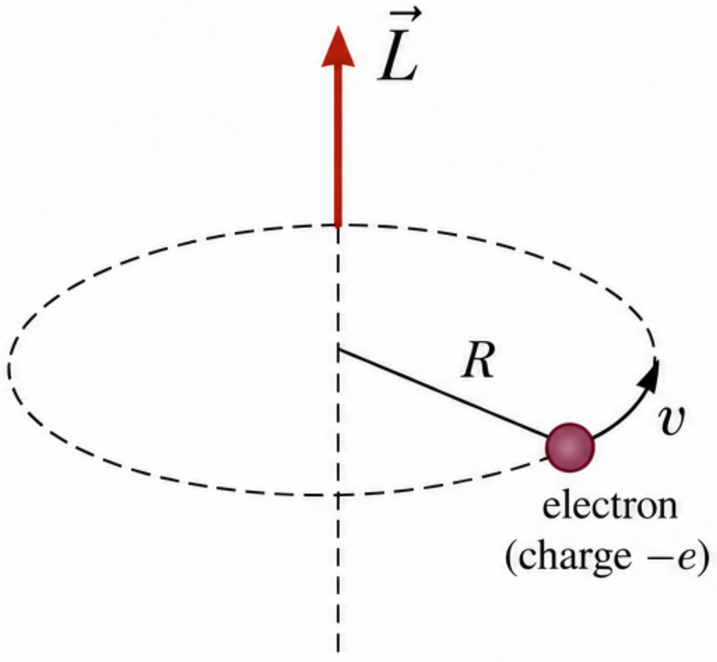
$$E'_{nj} = E_n + \Delta E_{nj}$$

Fine structure: lifting of degeneracy



Magnetic dipole moment from orbital motion

Electron in a circular orbit



A moving charge in a circular orbit constitutes a current loop and hence has a magnetic dipole moment.

1 Current due to orbital motion

The electron completes one revolution in a period $T = \dfrac{2\pi R}{v}$.

The equivalent current is

$$I = \frac{\text{charge}}{\text{period}} = \frac{-e}{T} = -\frac{ev}{2\pi R}.$$

2 Magnetic dipole moment of a current loop

The magnetic dipole moment of a current loop of area $A = \pi R^2$ is

$$\vec{\mu} = IA\,\hat{n} = -\frac{ev}{2\pi R}(\pi R^2)\hat{n} = -\frac{e}{2}Rv\hat{n}.$$

3 Relation to orbital angular momentum

The orbital angular momentum is

$$\vec{L} = m_e\vec{R} \times \vec{v}, \quad \text{with magnitude } |\vec{L}| = m_e Rv.$$

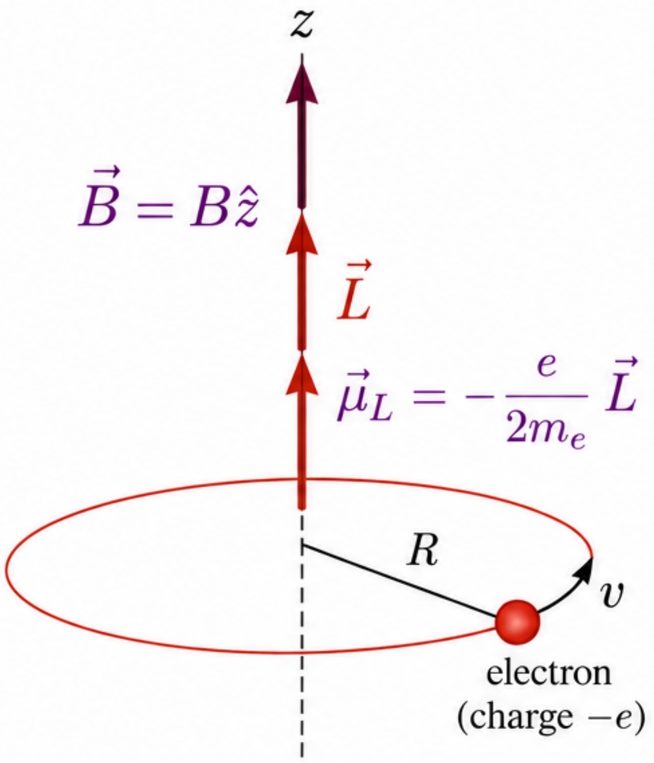
4 Magnetic moment in terms of $\vec{L}$

Hence,

$$\vec{\mu}_L = -\frac{e}{2m_e}\,\vec{L}.$$

Atom in a uniform external magnetic field: Zeeman effect (orbital)

Orbital magnetic moment in a uniform magnetic field



Orbital magnetic moment:

$$\vec{\mu}_L = -\frac{e}{2m_e}\vec{L}.$$

1 Interaction with an external magnetic field

The interaction energy is

$$U = -\vec{\mu}_L \cdot \vec{B} = \frac{e}{2m_e}\vec{L}\cdot\vec{B}.$$

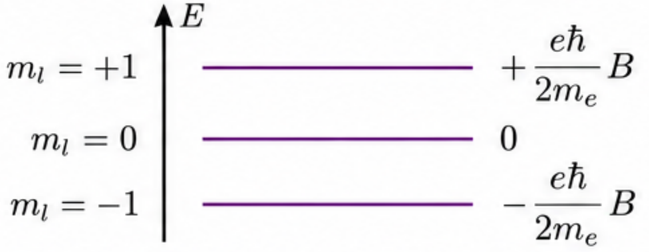
2 Energy shifts

For $\vec{B} = B\hat{z}$ and $L_z = m_l\hbar$ (with $m_l = -\ell, -\ell+1, \ldots, \ell$),

$$E_{ml} = \frac{e\hbar}{2m_e} B\, m_l, \qquad m_l = -\ell, -\ell+1, \ldots, \ell.$$

3 Zeeman splitting

The $(2\ell+1)$ degenerate levels split linearly with m_l. For $\ell = 1$ (p state):



4 Force and torque

The force on a magnetic dipole is $\vec{F} = -\nabla U$.
For a uniform $\vec{B}$, U is spatially constant $\Rightarrow$

$$\vec{F} = 0$$

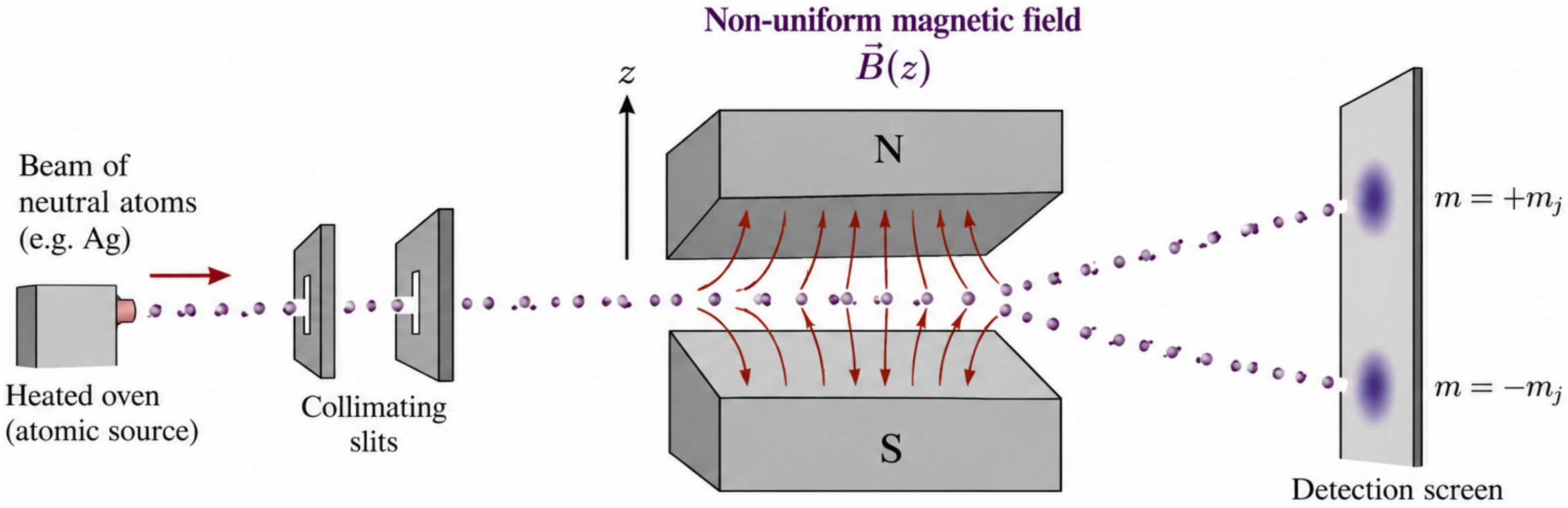
(no net translational force)

The torque is $\vec{\tau} = \vec{\mu}_L \times \vec{B}$, which is in general non–zero and tends to align $\vec{\mu}_L$ with $\vec{B}$.

$$\vec{\tau} \neq 0$$

(torque is non zero)

Stern Gerlach Experiment



Stern Gerlach experiment for l = 1

Non-uniform field and force

The potential energy of a magnetic dipole is

$$U = -\vec{\mu}_L \cdot \vec{B} = -\mu_{Lz} B(z).$$

The force along z is

$$F_z = -\frac{dU}{dz} = \mu_{Lz} \frac{dB}{dz}.$$

Since the field is stronger at the top, $\frac{dB}{dz} > 0$, **so the force is non-zero.**

Orbital magnetic moment and m_l

For an electron in an atomic orbital,

$$\vec{\mu}_L = -\frac{e}{2m_e}\vec{L}, \quad L_z = m_l \hbar,$$

so

$$\mu_{Lz} = -\frac{e\hbar}{2m_e} m_l.$$

$$F_z = -\frac{e\hbar}{2m_e} m_l \frac{dB}{dz}.$$

Deflection for $l = 1$ (three spots)

For $l = 1$, $m_l = -1, 0, +1$:

m_l	μ_{Lz}	F_z (for $dB/dz > 0$)	Deflection
$+1$	$-\frac{e\hbar}{2m_e}$	> 0	upward
0	0	0	no deflection
-1	$\frac{e\hbar}{2m_e}$	< 0	downward

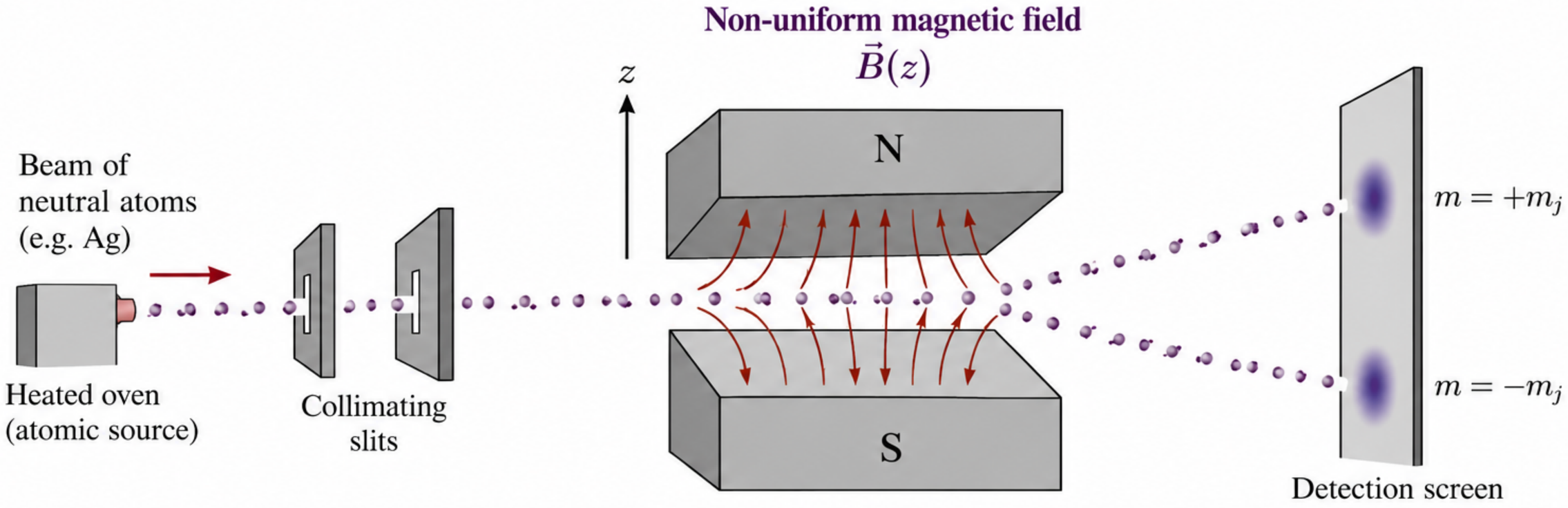
Thus, a beam of atoms with $l = 1$ is **expected to split into three spatially separated spots.**

The splitting we are talking about here is spitting seen on screen because some atoms are deflected upward and some downward. Splitting in zemen effect is in energy levels.

Stern Gerlach experiment for $l = 0$

If I send H atoms where electron are in ground state, that is orbital angular momentum is 0, we expect all atoms to be concentrate around centre.

Stern Gerlach Experiment



But we get two spots for l=0 electron state. Note that the spots are actually deflection of atom's trajectory. These must be due to the force we discussed two slides back.

Electron Spin

The electron possesses an additional, **intrinsic angular momentum S** called **spin**.

For an electron:

$$s = \frac{1}{2}, \qquad m_s = \pm\frac{1}{2}.$$

Hence the two Stern–Gerlach outcome.

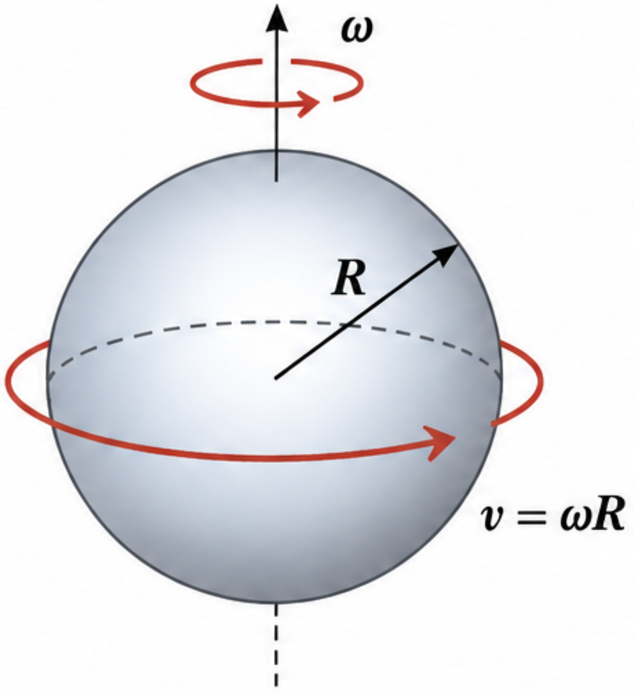
It will also follow similar algebra:

$$S^2 = s(s+1)\hbar^2, \qquad s = 0,1,2,\ldots,n-1$$

$$S_z = m_s\hbar, \qquad m_s = -s, -s+1,\ldots,0,\ldots,s-1,s$$

Suppose it really were spinning

Model: electron as a rotating sphere



1 **Angular momentum of a rotating object**

$$S \sim m_e R v$$ (order-of-magnitude estimate)

2 **Electron spin**

$$S \sim \frac{\hbar}{2}.$$

3 **Required surface velocity**

$$v \sim \frac{\hbar}{2 m_e R}.$$

4 **Take a typical electron size**

$$R \sim 10^{-15}\ \mathrm{m}$$ (nuclear scale, generous estimate)

5 **Resulting velocity**

$$v \sim \frac{\hbar}{2 m_e R} \sim 10^{11}\ \mathrm{m\,s^{-1}} \gg c.$$

6 **Can spin be ordinary mechanical rotation?**

No.

Electron Spin

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For an electron:

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Hence the two Stern–Gerlach outcome

Is ψ_{nlm} the complete description of an electron?

Stern–Gerlach says **no**.

The wavefunction should have

spatial wave function + spin degree of freedom

Electron Spin

Electron

↓

Quantum object

↓

$$s = \frac{1}{2}$$

So an electron in an atom now has TWO angular momenta, Land S. Should energy levels depend on how L and S are oriented relative to each other

End of Lecture 5